GlucoTwin

- A Personalized, Continuously Updating Digital Twin for Early Prediction of Glucose Excursions
- From Reactive Glucose Monitoring to Personalized Predictive Health.
- Vansh Deep Singh | IIIT Allahabad | B.Tech ECE, 2023-2027

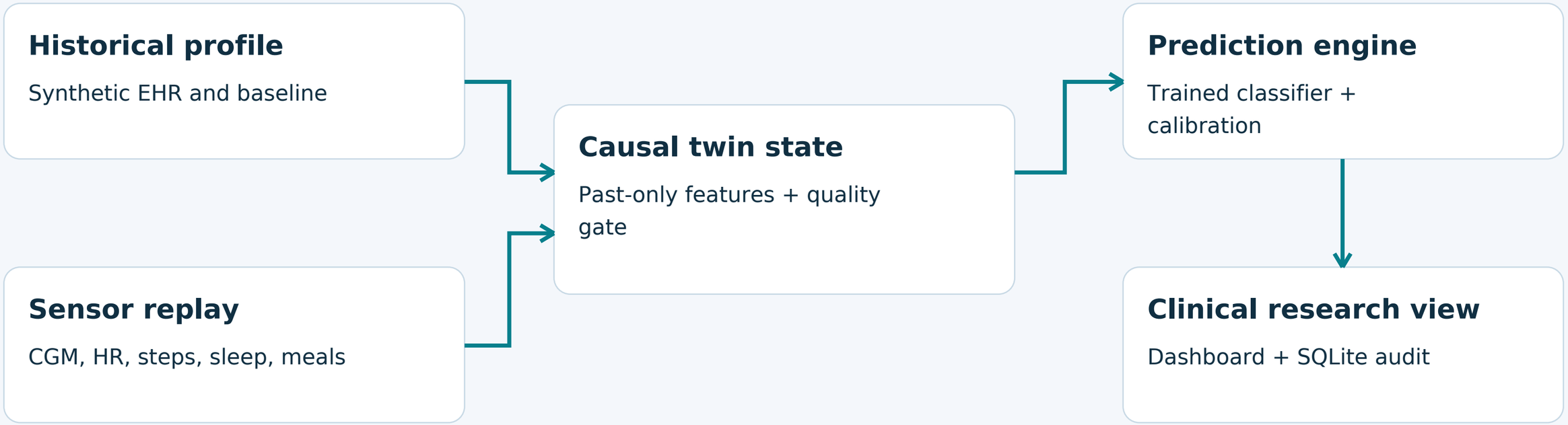
The problem: snapshots miss changing trajectories

- Historical measurements describe a baseline. Sensor streams describe what is changing.
- This prototype joins both streams to estimate a future high-glucose event.
- Research question: can recent observations and patient history predict the simulator-defined event?

A precise prediction contract

- At t: use only information available through t.
- Event: any glucose >180 mg/dL during t+5 to t+120 minutes.
- Require a complete future horizon to assign a training label. Report currently-below-threshold windows separately.

A continuously updating, patient-specific monitoring twin



Offline: simulator > labels > patient split > train > validate > calibrate > final test

Future observations are used only for labels and explicit retrospective evaluation.

Synthetic data with temporal structure

- 160 fictional patients. Seven days each. 322,560 sensor rows.
- Patient baselines, delayed meal peaks, activity-associated changes, circadian patterns and autocorrelated noise.
- Random and contiguous CGM dropouts. Hidden response parameters excluded from model features.

A trained model with patient-separated evaluation

- 88 training / 24 validation / 24 calibration / 24 test patients.
- Histogram gradient boosting: select leaf count using validation average precision.
- Separate sigmoid calibration. Fixed 0.50 alert cutoff. No test-set tuning.

Held-out performance on synthetic patients

Model	Precision	Recall	F1	PR-AUC
Fused model	0.873	0.819	0.845	0.950
Rule baseline	0.723	0.554	0.627	0.562
Dynamic-only	0.883	0.814	0.847	0.951

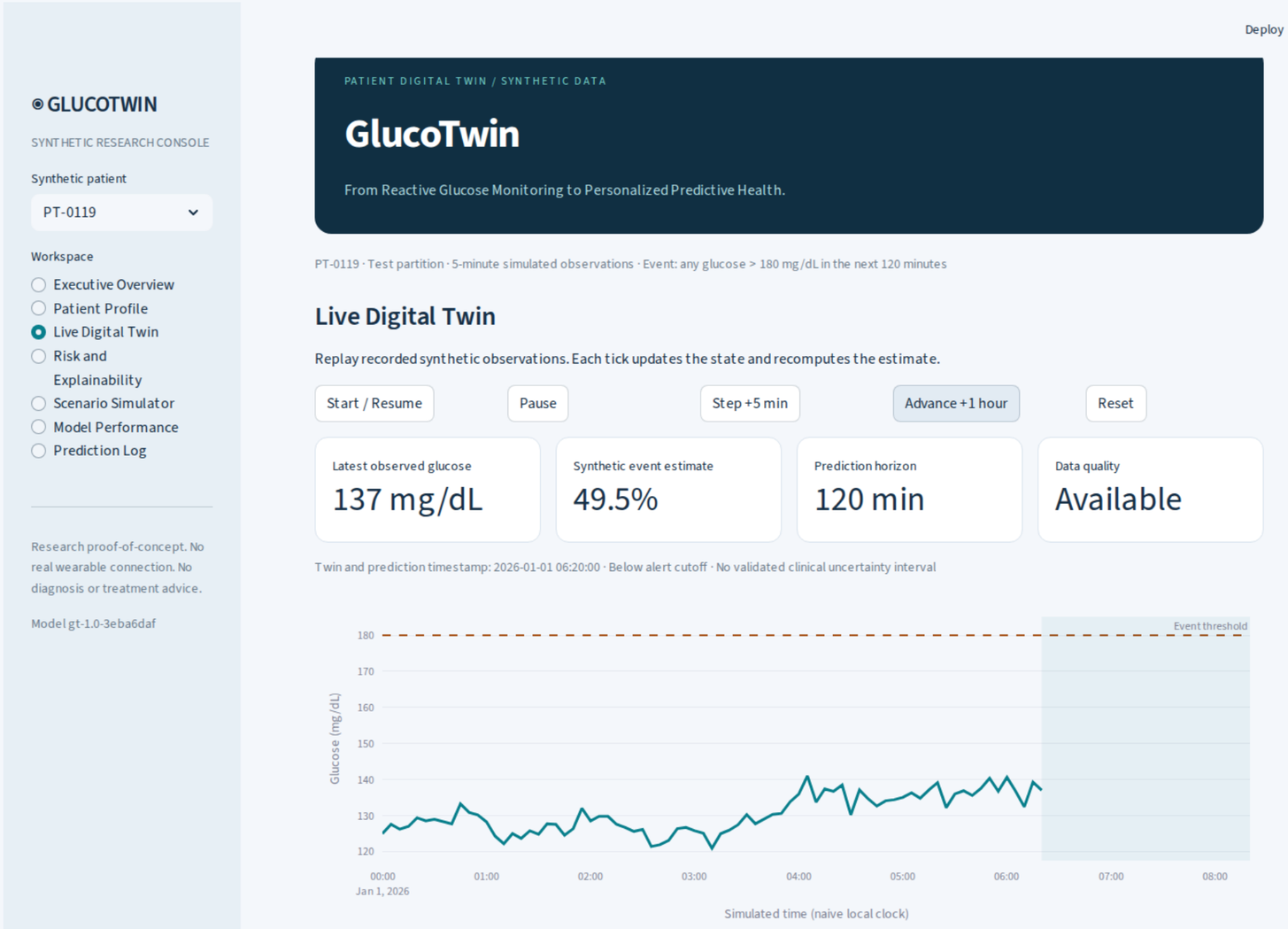
11,567 test windows from 24 held-out patients. PR-AUC = average precision.

Fused false-positive rate: 6.78%. Correlated windows are not independent alert episodes.

The result has important limits

- Dynamic-only F1: 0.847. Fused F1: 0.845. No demonstrated personalization benefit.
- Brier slightly worsened after calibration: 0.06867 to 0.06952.
- Below-threshold-at-t F1: 0.745. Main results include continuation of existing elevations.

The twin changes as new observations arrive



Start / pause / resume

Step / advance / reset

Past-only live view

Persistent audit log

Interpretation and scenario exploration

- Local sensitivity replaces one feature with its training median. It is non-additive and non-causal.
- Scenarios change a copied observed history, then recompute features and the estimate.
- A missing sensor interval withholds the estimate. No intervention or treatment advice.

Reproducible by design

- Seeded generator, pinned dependencies and bundled trained model.
- Shared feature code for training and replay. Persistent idempotent SQLite audit.
- Tests cover future-data isolation, labels, partitions, missingness, replay and dashboard navigation.

What the prototype establishes

- An end-to-end patient monitoring twin with real model training and evolving state.
- It predicts assumptions built into a synthetic simulator. It does not establish clinical validity.
- Next research step: independent licensed longitudinal data, patient-level uncertainty and prospective evaluation.

A transparent submission

- Runnable local application, source, trained model, metrics and documentation.
- Architecture and presentation PDFs, demonstration script and walkthrough materials.
- Public repository and hosted recording links must be verified before submission.